USED CARS PRICE ANALYSIS

Domain: Automotive Analytics

ExcellenC
Data Analytics

1. Introduction:

- ✓ The used car market is highly dynamic, with prices influenced by multiple factors such as brand reputation, vehicle age, fuel type, mileage, and financing options.
- ✓ Customers often struggle to identify cars that provide good quality at an affordable price due to unstructured and inconsistent market data.
- ✓ This project aims to analyze used car data collected from online platforms and convert raw listings into meaningful insights using data analytics techniques and interactive dashboards.

2. Problem Statement:

- ✓ The main objectives of this project are:
  - To analyze car price variations across different brands.
  - To identify best-value cars offering good quality at lower prices.
  - To study the impact of EMI, fuel type, mileage, on car pricing.
  - To visualize the distribution of cars by brand and company.
  - To support data-driven decision-making using dashboards.

3. Tools & Technologies:

- ✓ Python (Web Scrapping, Data cleaning and EDA), Power BI (Visualization & Reporting)

4. Data Collection:

- ✓ Data was collected from the **Cars24** website using **Python Selenium web scraping**.
- ✓ Key attributes include brand, model name, year, kilometers driven, fuel type, transmission, price, EMI, and city.
- ✓ Web scraping was chosen to ensure real-time and market-relevant data.

5. Methodology:

- ✓ This project follows an end-to-end data analytics pipeline:
  - Data collection using Selenium.
  - Data cleaning and preprocessing using pandas.
  - Feature engineering to derive business metrics.
  - Exploratory Data Analysis (EDA).
  - Dashboard development using Power BI.
- ✓ This structured approach ensures accuracy, interpretability, and scalability of insights.

6. Data cleaning and preparation:

- ✓ Raw scraped data contained inconsistencies and missing values. The following cleaning steps were performed:
  - Removed duplicate car listings.
  - Cleaned price values and standardized numeric formats.
  - Converted year, EMI, and kilometers into numeric fields.
  - Standardized fuel types and transmission categories.
  - Handled missing EMI values using brand-wise median imputation.
- ✓ These steps ensured data quality and analytical reliability.

7. Feature Engineering:

- ✓ New features were created to enhance analysis:
  - Calculated car_age, usage_intensity to identify the vehicle performance.
  - Value score was calculated using price, car_age and mileage to identify high-value score indicates better quality at lower cost.
  - Brand segmentation into Budget, Mainstream, and Luxury categories.

8. Exploratory Data Analysis:

- ✓ EDA was performed to understand relationships between variables:
  - Brand Vs price.
  - Fuel Type vs Median price.
  - EMI distribution across brands.
  - High value score models.
- ✓ These analyses helped identify pricing trends and consumer-friendly options.

9. Dashboard and Visualization:

- ✓ Interactive dashboards were created using Power BI with the following visuals:
  - ➢ Company vs Average Price.
  - ➢ Number of cars per brand.
  - ➢ Fuel type vs median price.
  - ➢ Top Value cars.
- ✓ Filters and slicers allow dynamic exploration by brand, fuel, and price range.

10. Key Insights:

- ✓ Premium brands show higher average prices but not always higher value scores.
- ✓ Some mainstream brands provide excellent quality at lower prices.

- ✓ Petrol and Diesel cars dominate the used car market.
- ✓ EMI values vary significantly across brands.

11. Challenges Faced:

- ➢ Dynamic website structure during web scraping.
- ➢ Inconsistent price formats (₹, lakh, commas).
- ➢ Missing EMI values.
- ➢ Duplicate listings across scroll loads.
- ➢ Designing a balanced value score metric.

- These challenges were resolved through robust cleaning logic and feature engineering.

12. Why this Methodology is Correct:

- ✓ This methodology follows industry-standard data analytics practices.
- ✓ Web scraping ensures real-world relevance, structured cleaning improves data quality, feature engineering enhances insight depth, and Power BI dashboards enable clear business communication.

13. Conclusion:

- ✓ This project successfully demonstrates how raw used car data can be transformed into actionable insights.
- ✓ The analysis helps identify price trends, best-value cars, and key market drivers, supporting informed decision-making for buyers and analysts.